10.3 Functions with Spring Cloud Function

Oleg Zhurakousky, project lead for Spring Cloud Function and Spring Cloud Stream, often asks conference audiences this question: Is there any business feature that you cannot define in terms of suppliers, functions, and consumers? It's an interesting and challenging question. Can you think of anything? Most software requirements can be expressed with functions.

Why use functions in the first place? They are a simple, uniform, and portable programming model that is a perfect fit for event-driven architectures, inherently based on these concepts.

Spring Cloud Function promotes the implementation of business logic via functions based on the standard interfaces introduced by Java 8: `Supplier`, `Function`, and `Consumer`.

- *Supplier*—A supplier is a function with only output, no input. It's also known as a *producer*, *publisher*, or *source*.
- *Function*—A function has both input and output. It's also known as a *processor*.
- *Consumer*—A consumer is a function with input but no output. It's also known as a *subscriber* or *sink*.

In this section, you'll learn how Spring Cloud Function works and how to implement business logic via functions.

10.3.1 Using the functional paradigm in Spring Cloud Function